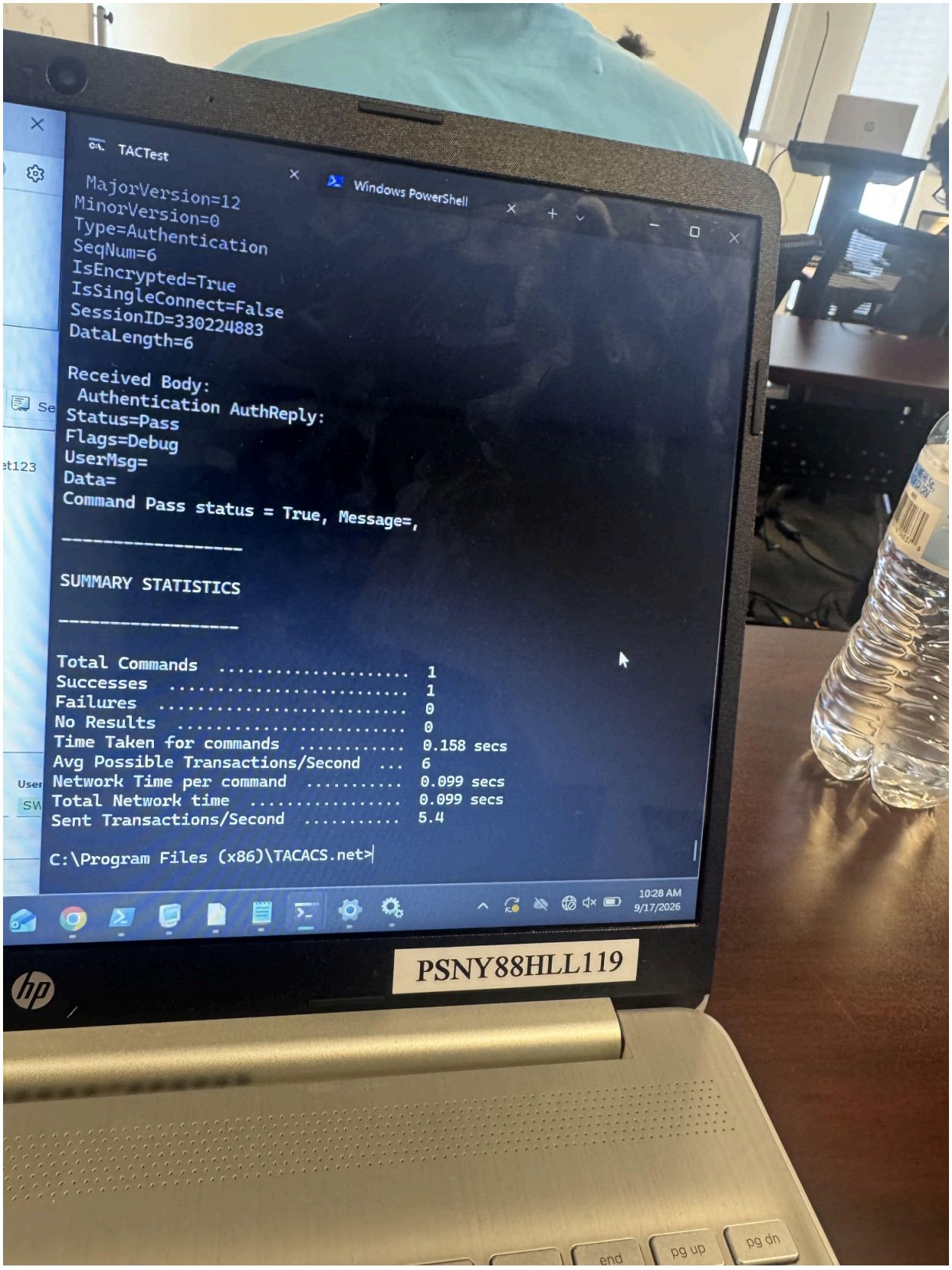
Cisco Secure Remote
Administration & AAA Lab

We first restarted the switch so we could start in Tera Term with a fresh configuration and avoid logging in before we began configuring. In Phase A, we set up the network settings we needed to configure the switch. We then went into configuration mode by typing (config term) so we could make changes and assign the switch a hostname, which made things easier. Next, we assigned VLAN 1 an IP address so the switch could be managed over the network. We then created and configured a default gateway so the switch could communicate with other devices on the network. For the last part of this step, we used the commands "show ip interface brief" and "ping" to verify that the connection was working properly. For phase B, we first checked the switch to see if it supported SSH, then we configured a domain name, created a local administrator account, and provided encryption by generating RSA keys. We then enabled SSH version 2 and wanted SSH connections to be able to use the local login information, so we configured VTY lines. After that, we made sure that SSH was configured properly and opened a second terminal so that we could connect to the switch remotely using the management IP address and TCP port 22. For a backup, we kept the original connection open just in case anything went wrong. Lastly, we tested to make sure that the SSH connection and remote management were working correctly.

For Phase C, we basically set up the Windows 11 computer to be our AAA server. First, we wanted the switch to always know where to locate the server, so we gave it a stable IP address. Then, we created firewall rules to allow RADIUS traffic on UDP port 1812 and TACACS+ on TCP port 49. For RADIUS authentication, we installed TekRADIUS, and for handling TACACS+, we installed TACACS.net. Finally, we made sure everything was running and that the ports were listening for connections. For the first step in phase D, for the Cisco switch to be able to use the AAA server, we configured RADIUS. We added the switch to TekRADIUS as a RADIUS client to be able to use its IP address and shared secrets. We wanted to test the TekRaduis service, so we created a test user to make sure everything was running properly on UDP port 1812. Next, on the Cisco switch, we enabled AAA and configured it to send a request to log in to the RADIUS server. RADIUS was set up as the first authentication method. The switches' local account serves as a backup if the server stops working. For the last step, we ran a test to see if RADIUS would reject a login when someone logs in successfully or when a password is incorrect, and we also tested the local fallback by temporarily stopping RADIUS.

In Phase E, we configured TACACS.net onto the Windows 11 AAA server by adding users and permissions and rules for the authentication settings. We made sure that the configuration was working and that TACACS+ was running on TCP port 49. Then, we configured the switch with the R+TACACS+ IP address from the server and the shared secrets as well. We wanted the switch to use TACACS+ for the first authentication, so we changed it, and the local administrator account would be used as a backup. Finally, we tested both the TACACS+ and the local backup login to make sure they were working properly. For Phase F, we used TACACS+ and our

executive privileges to control what users are allowed to do after they login and keep records of their sessions.


TACTest
Windows PowerShell
MajorVersion=12
MinorVersion=0
Type=Authentication
SeqNum=6
IsEncrypted=True
IsSingleConnect=False
SessionID=330224883
DataLength=6
Received Body:
Authentication AuthReply:
Status=Pass
Flags=Debug
UserMsg=
Data=
Command Pass status = True, Message=,
SUMMARY STATISTICS
Total Commands 1
Successes 1
Failures 0
No Results 0
Time Taken for commands ..... 0.158 secs
Avg Possible Transactions/Second ... 6
Network Time per command ..... 0.099 secs
Total Network time ..... 0.099 secs
Sent Transactions/Second 5.4
C:\Program Files (x86)\TACACS.net>
10:28 AM
9/17/2026
PSNY88HLL119
end
pg up
pg dn